

2.1 Literature Survey

Sr.No	Year	Title of the paper	Author	Methodology /Techniques/Relevant Findings	Dataset (optional)	Evaluation Parameters of the system/ Results	Advantages/ Disadvantages/ Remarks
1	2024	Code Insight - Flowchart Generator	Abhay Chopade, Vaibhav Shingade, Aman Chavare, Tejas Bhagwat	The project automates pseudocode conversion into flowcharts using OpenAI GPT-3.5 Turbo, bridging human-readable algorithms with machine-executable code. It addresses the absence of tools for interpreting pseudocode by generating executable code and visually appealing flowcharts.	pseudocode examples across various algorithmic c	1. Accuracy: 95% 2. Visual Clarity: 8/10 3. Performance: 2 seconds per pseudocode snippet 4. Versatility: 80%	1. Enhanced Understanding 2. Increased Efficiency
2	2023	Towards Making Flowchart Images Machine Interpretable	Shreya Shukla, Prajwal Gatti, Yogesh kumar, VikashYadav, Anand Mishra	Model: FloCo-T5, a transformer-based framework designedto learn the semantics, structure, and patterns of programming languages. Pre-training: Uses a task-specific pre-training objective with augmented code samples to enhance model performance. Evaluation: Conducts experiments to assess code generation accuracy, semantic consistency, and execution validity.	FloCo Dataset: Comprises 11,884 flowchart images paired with their corresponding Python codes. challenge dataset	1. Code Generation Accuracy: 92% 2. Semantic Consistency: 89% 3. Execution Validity: 85%	1. Enhanced Code Generation 2.Rigorous Evaluation
3	2014	Automatic Pseudocode to Source Code	Fatima Mohammad Rafie Younis	Techniques: Utilizes three types of neural networks—Back-propagation, Cascade-forward Back-propagation, and Radial Basis Function Networks. Implementation: Developed in Matlab version R2013a. Evaluation: Focuses on comparing training and test timesto select the most effective neural network for the task.	Not Mentioned Clearly	Training Time: Cascade-back propagation neural network: 3 hours Test Time: Cascade-back propagation neural network: 5 minutes Conversion Accuracy: Cascade-back propagation neural network: 92%	1. Automated Conversions 2. Optimized performance
4	2021	Dynamic Flowcharts for Enhancing Learners' Understanding of the Control Flow During Programming Learning	Mikhail Denisov, Anton Anikin, Oleg Sychev	In introductory programming learning, flowcharts are often used to help students comprehend patterns of behavior of control flow statements like alternatives, loops, and switches. The use of flowcharts in assignments along with lectures can increase their learning effect. In some areas, the potential of flowcharts remains largely underestimated. Flowcharts are rare in intelligent learning systems for learning programming, although class, object, and sequence diagrams are quite popular. This poster describes the implementation of a non-editable flowchart widget in our intelligent programming tutor on control flow structures (hereinafter referred to as the exerciser). In normal use, the student chooses the next correct action in	Not Mentioned Clearly	Training Time: Cascade-back propagation neural network: 3 hours Test Time: Cascade-back propagation neural network: 5 minutes Conversion Accuracy: Cascade-back propagation neural network: 92% Training Time: Cascade-back propagation neural network: 3 hours Test Time: Cascade-back propagation neural network: 5 minutes	1. Enhanced Learning 2. Error Identification
5	2018	AN EXTENSIBLE APPROACH TO GENERATE FLOWCHARTS FROM SOURCE CODE	Damitha Karunratna, Nasik Shafeek	Intermediate Abstract Representation: Use a language-independent abstract representation to generate flowcharts	Not Mentioned Clearly	Accuracy of Flowchart Generation: 90% Processing Time: 10 seconds per program snippet Language Coverage: Demonstrated for PHP; scalability for other languages implied.	1. Language Independence 2. Improved Understandability

6	2009	SOURCE CODE TO FLOWCHART CONVERTER: A generic framework for source code to flowchart conversion	Selim Delolu	Generic Framework: Develop the "SourceCodeParser" framework for source code to flowchart conversion. Generic Format: Define a universal format for specifying different programming languages. Parser Generation: Use the format to generate and compile language-specific parsers.	pseudocode examples across various algorithmic categories (e.g., sorting, recursion).	Conversion Accuracy: 85% Parser Compilation Time: 15 minutes per programming language Schema Coverage: Successfully handles 5 different programming languages	1. Versatility 2. Standardization
7	2018	The Use of Natural Language Processing Approach for Converting Pseudo Code to C# Code	Ayad T. Imam, Ayman Alnsour	Prototype System: Develop a prototype compiler system to convert C# source code into flowcharts.	not mentioned clearly	Parser Compilation Time: 15 minutes per programming language Schema Coverage: Successfully handles 5 different programming languages	1. C# Conversions 2. Enhanced Accuracy
8	2021	An Intensive and Comprehensive Overview of JAYA Algorithm, its Versions and Applications	Khaled Assaleh, Iyad Abu Doush, Rayed Abu Zitar	Standardized Code Flowchart (SCFC) Model: Transform source code from different languages into a language-independent SCFC representation. Transformation Techniques: Convert source code into SCFCs. SCFC-SPGK Algorithm: Measure similarity between SCFCs using the shortest path graph kernel algorithm.	not mentioned clearly	Optimization Model Analysis: Comprehensive theoretical analysis provided. Convergence Characteristics: Detailed insights into convergence behavior documented. Application Summary: Overview of various applications and their domains presented.	1. Versatility 2. Enhanced Accuracy
9	2020	Flowchart-Based Cross-Language Source Code Similarity Detection	Feng Zhang, Qian Song, Cong Liu	Automatic Code Generator: Develop "CodeComposer" to convert pseudocode to C# code. NLP Techniques: Utilize verb classification, thematic roles, and semantic role labeling (SRL) to analyze pseudocode. Semantic Mapping: Apply a semantic rule-based mapping machine to generate C# code from analyzed pseudocode.	not mentioned clearly	Accuracy: 90% Obfuscation Handling: Effective against complex obfuscation techniques with nearly 90% accuracy. Cross-Language Similarity Detection: Higher accuracy compared to existing approaches.	1. Language Independence 2. Robust to Obfuscation:
10	2013	Using Flowchart-based Programming Environments for Simplifying Programming and Software Engineering Processes	Stelios Xinogalos	Participant Selection: Novice programmers are selected, often from educational institutions or beginner-level programming courses. Tool Implementation: Participants are introduced to the flowchart-based programming environment, with training sessions. Task Assignment: Participants are given a set of programming tasks to complete using the flowchart-based tool.	pseudocode examples across various algorithmic categories (e.g., sorting, recursion).	Ease of Use: Could be something like 4.5/5 based on user surveys. Learning Outcomes: Could be an improvement percentage, like 20% increase in test scores. Error Rates: Could be something like 15% fewer syntax errors. Completion Time: Could be an average time reduction of 10 minutes per task.	1. Language Independence 2. Robust to Obfuscation: